Intent and sentiment aware IR

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Overview

Introduction

	Report	Baseline 1	Baseline 2	Baseline 3	Advanced Exp. 1	Advanced Exp. 2
Dario	●	●				
Linda	●			●	●	●
Michele	●		●		●	●

Problem Statement

01 Non-expert difficulty of accessing financial terms and concepts

- 1.2 Low financial literacy in most EU countries *
- 1.3 Altered sentiment perception of financial terms

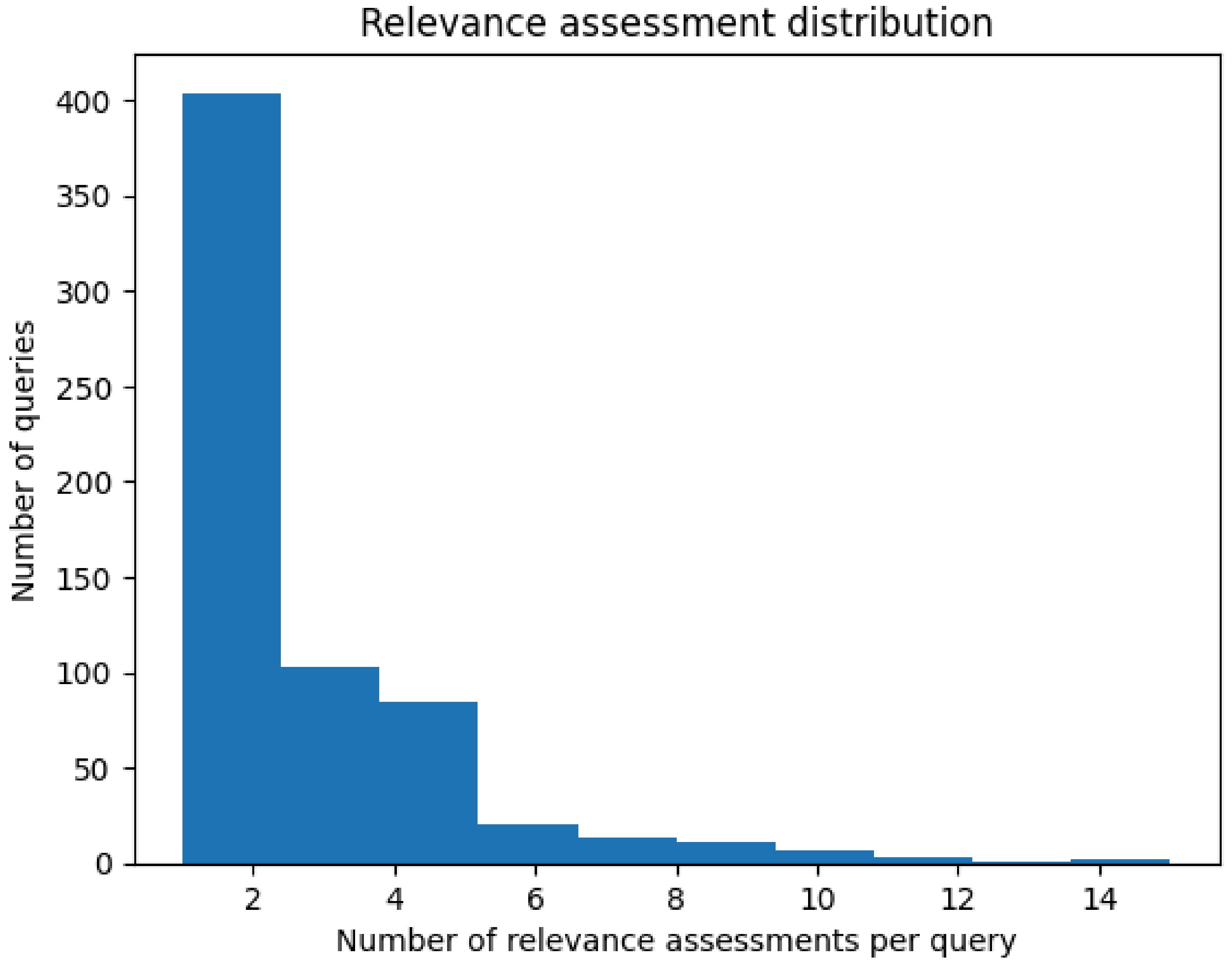
02 Diffusion of overly specific and one-sided retrieval systems

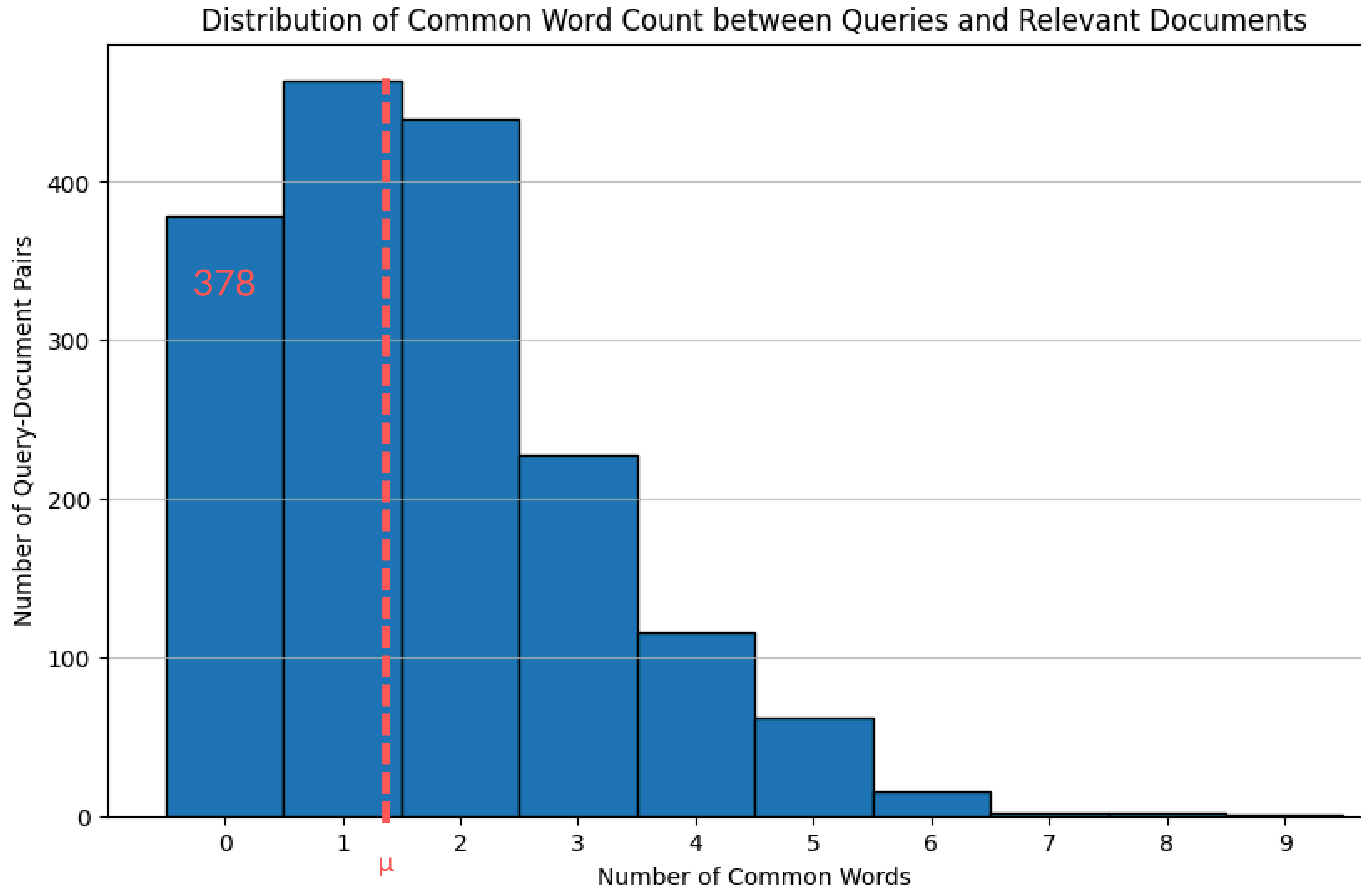
03 Diffusion of intent and sentiment unaware systems

*EU 2023 Study: https://europa.eu/eurobarometer/surveys/detail/2953

EDA

Description	Value
#Queries	1296
Query avg length	10.94
MAX length of queries	31
MIN length of queries	2
#Queries shorter than 5 words in the whole collection	29
#Queries with more than 1 relevant doc	428
Mean number of queries relevance assessment	2.63
#Queries with a number inside	73
#Queries with a sentiment word	111





Description	Value
#Docs	57638
Docs avg length	132.90
MAX length of documents	2973.00
MIN length of documents	0
#Docs shorter than 5 words in the whole collection	73
#Short documents relevant to some queries	2
#Docs with emojis/non latin alphabet characters	49
#Docs with non standard money quantities (e.g 5k)	3747
#Docs containing mentions or hashtags	1834
#Docs with links	5946
#Docs with extended (>5 words) quoted text	14386

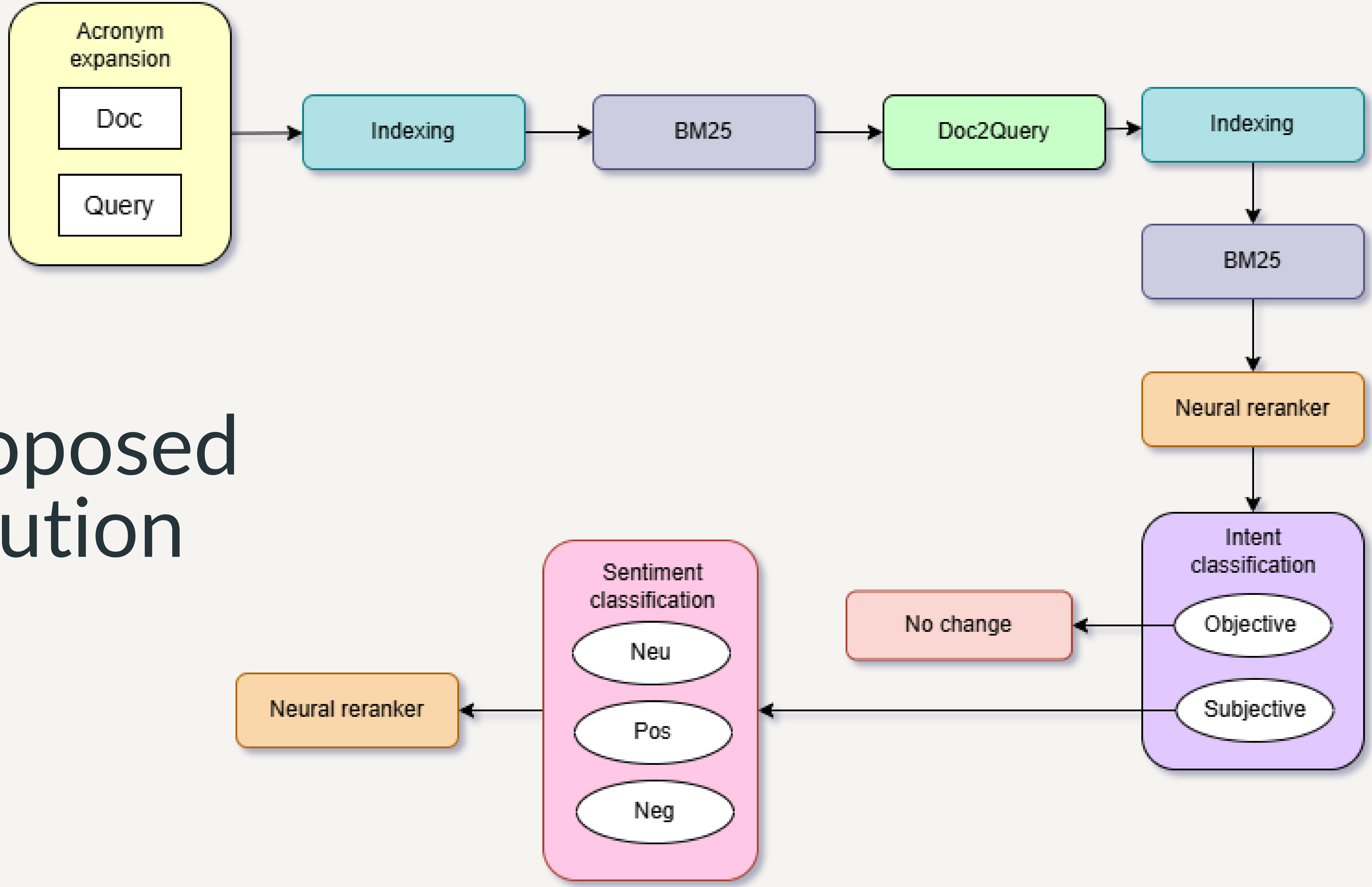
Research Objectives

- Investigate the effect on IR models of financially informed preprocessing steps
- Understand the effect of query rewriting to expand the expression of the information need of the users. And the document expansion to improve IR models performances
- Evaluate the effect of sentiment and intent classifiers to generate a more varied final ranking list

Hypotheses

1. Increasing the number of matching words between relevant docs & query improves the performances of statistics based models such as BM25

Proposed solution



Baseline methods

1

- Basic preprocessing
- Non alphabetical symbols removal

2

- Synthatic expansion
- Acronym expansion
- Morphological expansion
- Query rewriting

3

- Chunking
- Document filtering (size)
- Document filtering (language)

Results

Baseline:

	name	map	P.1	P.5	P.10	recall.5	recall.10	ndcg_cut_5	ndcg_cut_10
0	BM25	0.206431	0.231481	0.107407	0.070216	0.247984	0.313926	0.227734	0.250566

Advanced experiments:

	name	map	P.1	P.5	P.10	recall.5	recall.10	ndcg_cut_5	ndcg_cut_10
0	BM25	0.207961	0.233025	0.108642	0.072068	0.249016	0.327327	0.230265	0.257055
1	E1	0.285493	0.344136	0.154630	0.093981	0.352451	0.412465	0.328618	0.347721
2	E2	0.279608	0.336420	0.151852	0.093673	0.345472	0.410921	0.321310	0.342616

Main outcomes

```
initial_ranking = results_E1[results_E1['qid'] == '10136'].head()
display(initial_ranking)
```

	qid	docid	docno	rank	score	query	text
37500	10136	4924	51337	0	0.933398	How to minimise the risk of a reduction in purchase power in case of Brexit for money held in a bank account?	By your logic, if a loan of £100 is new money dilutes your purchasing power, then the repayment of £110 is a reduction of the money supply that increases your purchasing power. Indeed, ultimately the increase in purchasing power upon repayment is greater than the initial reduction, so you are 'better off' every time a loan is made and successfully repaid. The effect on you is tiny, but the collective benefit you get from all the loans being repaid with interest is more or less equivalent to the purchasing power reduction of the loans that are never repaid. Therefore you do not lose out and are indeed compensated (in a tiny way) for the tiny risk you incurred. The bank incurs a substantial risk and is thus compensated in a substantial way.
37504	10136	44878	466950	4	-2.128047	How to minimise the risk of a reduction in purchase power in case of Brexit for money held in a bank account?	"Having savings only in your home currency is relatively 'low risk' compared with other types of 'low diversification'. This is because, in a simple case, your future cash outflows will be in your home currency, so if the GBP fluctuates in value, it will (theoretically) still buy you the same goods at home. In this way, keeping your savings in the same currency as your future expenditures creates a natural hedge against currency fluctuation. This gets complicated for goods imported from other countries, where base price fluctuates based on a foreign currency, or for situations where you expect to incur significant foreign currency expenditures (retirement elsewhere, etc.). In such cases, you no longer have certainty that your future expenditures will be based on the GBP, and saving money in other currencies may make more sense. In many circumstances, 'diversification' of the currency of your savings may actually increase your risk, not decrease it. Be sure you are doing this for a specific reason, with a specific strategy, and not just to generally 'spread your money around'. Even in case of a Brexit, consider: what would you do with a bank account full of USD? If the answer is ""Convert it back to GBP when needed (in 6 months, 5 years, 30, etc.), to buy British goods"", then I wouldn't call this a way to reduce your risk. Instead, I would call it a type of investment, with its own set of risks associated."
37533	10136	9241	95246	33	-2.829449	How to minimise the risk of a reduction in purchase power in case of Brexit for money held in a bank account?	These products are real, but they aren't risk free: 1) The bank could go under in that time. (Are the investments FDIC insured?) 2) Your money is locked up for 5 years, probably with either no way to get it back out or a stiff penalty for early withdrawal, so you risk having a better investment opportunity come along and not having the liquidity to take advantage of it. 3) If the market does go down and you get 100% of your principal back, the endless ratchet of inflation practically guarantees that $10K will be worth less 5 years from now than it is today, so you risk losing purchasing power even if you're not losing any nominal quantity of money. It's still a fairly low-risk investment option, particularly if it's tied to something that you have reason to believe will increase in value significantly faster than inflation in the next 5 years.

```
new_ranking = results_E2[results_E2['qid'] == '10136'].head()
display(new_ranking)
```

	qid	docid	docno	rank	score	query	text	sentiment
400	10136	44878	466950	4	100	How to minimise the risk of a reduction in purchase power in case of Brexit for money held in a bank account?	"Having savings only in your home currency is relatively 'low risk' compared with other types of 'low diversification'. This is because, in a simple case, your future cash outflows will be in your home currency, so if the GBP fluctuates in value, it will (theoretically) still buy you the same goods at home. In this way, keeping your savings in the same currency as your future expenditures creates a natural hedge against currency fluctuation. This gets complicated for goods imported from other countries, where base price fluctuates based on a foreign currency, or for situations where you expect to incur significant foreign currency expenditures (retirement elsewhere, etc.). In such cases, you no longer have certainty that your future expenditures will be based on the GBP, and saving money in other currencies may make more sense. In many circumstances, 'diversification' of the currency of your savings may actually increase your risk, not decrease it. Be sure you are doing this for a specific reason, with a specific strategy, and not just to generally 'spread your money around'. Even in case of a Brexit, consider: what would you do with a bank account full of USD? If the answer is ""Convert it back to GBP when needed (in 6 months, 5 years, 30, etc.), to buy British goods"", then I wouldn't call this a way to reduce your risk. Instead, I would call it a type of investment, with its own set of risks associated."	neutral
401	10136	4924	51337	0	99	How to minimise the risk of a reduction in purchase power in case of Brexit for money held in a bank account?	By your logic, if a loan of £100 is new money dilutes your purchasing power, then the repayment of £110 is a reduction of the money supply that increases your purchasing power. Indeed, ultimately the increase in purchasing power upon repayment is greater than the initial reduction, so you are 'better off' every time a loan is made and successfully repaid. The effect on you is tiny, but the collective benefit you get from all the loans being repaid with interest is more or less equivalent to the purchasing power reduction of the loans that are never repaid. Therefore you do not lose out and are indeed compensated (in a tiny way) for the tiny risk you incurred. The bank incurs a substantial risk and is thus compensated in a substantial way.	positive
402	10136	29483	304500	9	98	How to minimise the risk of a reduction in purchase power in case of	Behind paywall, so copy/paste of article below: > Brexit will push up costs for banks by as much as 4 per cent and their capital requirements will rise up to 30 per cent, according to the most detailed assessment yet of what Britain's departure from the EU (european union) means for the sector. > The findings by consultants Oliver Wyman will make grim reading for its bank clients, many of which are struggling with low profitability. They come a day after HSBC (hongkong and shanghai banking corporation) became the first lender to put a price tag on Brexit, saying the immediate disruption would cost it $200m-$300m. > Stuart Gulliver, chief executive of HSBC (hongkong and shanghai banking corporation), said $1bn of revenue in its global banking and markets unit would be put "at risk" by Brexit. But he said it planned protect this revenue by moving up to 1,000 of its 6,000 UK (united kingdom) investment banking jobs to France. > The pace of announcements about banks' Brexit plans has picked up in recent weeks, partly because of pressure from the Bank of England for them to submit their plans for coping with the "worst-case scenario" of a hard Brexit, severing access to EU (european union) clients. The UK (united kingdom) is set to leave the EU (european union) in March 2019. > Such plans are expected to cause duplication of resources and capital for large banks in Europe, Oliver Wyman warned. It said this may cause some banks to abandon some European activities altogether and shift resources to the US (united states) and Asia. > "At the moment what everyone is doing is planning to be able to continue doing what they already do after a hard Brexit," said Matthew Austen, head of European corporate and institutional banking at Oliver Wyman. > "Once you have done that, if you have a strong performance in the US (united states) or Asia, then that is when you start to look at the post-Brexit foundations and it will prompt you to look at what the right business mix is," he said. > The consultancy, which has access to detailed figures on almost every bank from the benchmarking work it does for the sector, estimated that 2 percentage points would be knocked off wholesale banks' return on equity in Europe because of the disruption. > Wholesale banks — which	negative

Lessons learned

01 What we think might help, doesn't always help → intuition might seem reasonable but has to be adapted to the specific data (e.g query rewriting)

02 Intent - sentiment trade off + quality of initial ranking matters → we have multiple compounding steps, the errors made in one step propagates in the others

03 There is always a feeling that something else has to be tried out, maybe that will be the key finding that will make a huge difference, so it becomes a never ending project; we had to learn when to make ourselves stop

Limitations

01 The attempt to induce semantic understanding in non-neural models doesn’t reflect in practical improvement of the model

02 We tried to implicitly implement diversity through diversification of sentiment of subjective query/doc but we don’t have a metric to evalute how novel and diverse the new raking is

Use of AI

01 Generation of the expansion of the acronyms absent in the glossary

02 General questions of brainstorming on possible usefulness of discussed pipelines

03 Explain errors

04 Help to quicly implement random ideas to check if they were viable

Thank You

Queries for the team?